

Removing Completeness from the Metric $C_b(K)$ Invariant-Subspace Classification

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Status

This is a partial/theorem-adjacent result for Cheng–Fang–Jiang, *On the third problem of Halmos on Banach spaces*, arXiv:2203.14670v2 [1]. It does not solve Halmos’s third problem for invertible operators. It strengthens the final $C(K)$ -classification in the source paper by removing the assumption that the metric space K is complete.

Source Passage

The source paper writes $C(K)$ for the Banach space of bounded continuous complex-valued functions on a metric space K , with the supremum norm. Theorem 16.1 assumes that K is complete:

16. PROOF OF THEOREM 1.9

Based on another beautiful result of Read [45] that an infinite dimensional separable complex Banach space X of the form $X = c_0 \oplus Y$ for some separable Y always admits an operator $T \in \mathfrak{B}(X)$ without any non-trivial invariant subspace, and the Sobczyk theorem [48] that c_0 is 2-complemented in every separable Banach space geometrically containing it, in this section, we show the following theorem.

Theorem 16.1. *Let K be a complete metric space such that the space $C(K)$ of all bounded continuous complex valued functions is infinite dimensional. Then a sufficient and necessary condition for $C(K)$ admitting an operator $T \in \mathfrak{B}(C(K))$ without any nontrivial subspace is that the metric space K is compact.*

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In the proof tail, completeness is used only in the noncompact case:

Then it is easy to observe that $C_0(K)$ is a closed subspace of $C(K)$ isometric to c_0 . Therefore, again by the Sobczyk theorem [48], c_0 is 2-complemented in $C(K)$. Therefore, the sufficiency follows from the previously mentioned theorem of Read [45].

Necessity. Note that every bounded linear operator on a nonseparable Banach space admits a nontrivial invariant subspace. Since K is a complete metric space, and since that there is $T \in \mathfrak{B}(C(K))$ without any nontrivial subspace, $C(K)$ must be separable and infinite dimensional. Therefore, the cardinality of K is infinite. If K is not compact, then it is not difficult to observe that ℓ_∞ can be linearly isometrically embedded into $C(K)$. This contradicts to that $C(K)$ is separable.

□

Claim

Let K be a metric space and let $C_b(K)$ denote the Banach space of bounded continuous complex-valued functions on K , with the supremum norm. Assume $C_b(K)$ is infinite-dimensional. Then $C_b(K)$ admits a bounded linear operator without nontrivial closed invariant subspaces if and only if K is compact.

Proof Intuition

The compact side needs no new work: compact metric spaces are complete, so the source theorem applies directly. The only issue is whether a noncomplete, noncompact metric space might evade the source paper's necessity argument. It cannot. Every noncompact metric space contains a closed discrete infinite sequence. Restricting bounded continuous functions to that sequence gives a quotient map from $C_b(K)$ onto ℓ_∞ , by Tietze extension. Thus $C_b(K)$ is nonseparable, and on a nonseparable Banach space every operator has a nontrivial invariant subspace: the closed orbit generated by one nonzero vector is separable.

Proof

We use two elementary facts.

Lemma 1. *Let X be a nonseparable Banach space. Then every $T \in \mathcal{B}(X)$ has a nontrivial closed invariant subspace.*

Proof. Choose $0 \neq x \in X$. The closed linear span

$$M = \overline{\text{span}}\{x, Tx, T^2x, \dots\}$$

is a nonzero closed T -invariant subspace. It is separable. Since X is nonseparable, $M \neq X$. □

Lemma 2. *If K is a noncompact metric space, then $C_b(K)$ is nonseparable.*

Proof. Since metric compactness is equivalent to sequential compactness, there is a sequence (x_n) in K with no convergent subsequence. Passing to a subsequence if needed, assume the x_n 's are distinct, and put $D = \{x_n : n \geq 1\}$.

The set D is closed and discrete in K . Indeed, any accumulation point of D would be the limit of a subsequence of (x_n) . Hence every bounded function on D is continuous, so $C_b(D)$ is isometric to ℓ_∞ . The restriction operator

$$R : C_b(K) \longrightarrow C_b(D), \quad Rf = f|_D,$$

is bounded and linear. It is onto by the Tietze extension theorem, because metric spaces are normal and D is closed. Therefore ℓ_∞ is a continuous linear image of $C_b(K)$. If $C_b(K)$ were separable, then ℓ_∞ would be separable, a contradiction. $\square$

Now suppose first that K is compact and $C_b(K)$ is infinite-dimensional. Then K is a complete metric space, and Theorem 16.1 of [1] gives a bounded operator on $C_b(K) = C(K)$ without nontrivial closed invariant subspaces.

Conversely, suppose that K is not compact. By Lemma 2, $C_b(K)$ is nonseparable. Lemma 1 then implies that every bounded operator on $C_b(K)$ has a nontrivial closed invariant subspace. So no bounded operator without nontrivial closed invariant subspaces exists on $C_b(K)$.

This proves the equivalence for arbitrary metric spaces K .

Verification and Limitations

The proof is independent of the difficult Read construction except through the compact-metric sufficiency already proved in the source paper. The new necessity argument uses only the standard facts that noncompact metric spaces are not sequentially compact, metric spaces are normal, and Tietze extension holds for bounded scalar-valued continuous functions on closed subsets.

This packet does not answer the source paper's main Halmos Problem 1.2. It only removes an unnecessary completeness assumption from the source paper's final $C(K)$ theorem.

Local run indexes were searched for the arXiv id and keywords **C(K)**, **C.b(K)**, **complete metric**, **bounded continuous**, and invariant-subspace terms. Web searches were run for the Halmos third problem and for $C_b(X)$ noncompact metric separability variants. No duplicate run packet or explicit later literature answer for this exact strengthening was found. Since the new step is standard-topology folklore-level, human review should treat the novelty claim conservatively.

Human Review Recommendation

Check that the source paper's $C(K)$ notation indeed means $C_b(K)$ for noncompact metric spaces, as stated in Theorem 16.1. Then verify the restriction/Tietze quotient argument in Lemma 2. If accepted, the source theorem can be restated for all metric spaces.

References

- [1] Lixin Cheng, Junsheng Fang, Chunlan Jiang, *On the third problem of Halmos on Banach spaces*, arXiv:2203.14670v2, 2022.
- [2] Heinrich Tietze, *Ueber Funktionen, die auf einer abgeschlossenen Menge stetig sind*, Journal fuer die reine und angewandte Mathematik **145** (1915), 9–14.